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PALEOMAGNETIC INVESTIGATIONS IN THE THÜRINGER FOREST (G.D.R.)

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ABSTRACT

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In the Thüringer Forest, the whole "Rotliegende" is well exposed, and was chosen as beginning of a Permian traverse via Czechoslovakia, Austria into Italy. Rockmagnetic experiments established haematite as main carrier of the NRM, except a few localities, where magnetite was found to be present. Sediments and volcanites show consistent ChRM directions. The comparison with KrŠ's (1978) compilation proves a good Permian pole position.

INTRODUCTION

The possibility of using paleomagnetic methods to complete and refine the stratigraphic time-scale is well known and generally accepted. The great advantage arises especially in horizons where the lack of fossils makes exact dating difficult. On this basis a project was started to compare the Permian (Rotliegende) of the Thüringer Forest (Th.F.) with the magnetostratigraphy of the European Permian (Dachroth, 1976) (see Fig. 2).

These data of the Th.F. are thought to be the starting point of a geotraverse running from the Th.F. through Czechoslovakia (Boskoviče) into Austria (Permian of Zöbing) and further into the Rotliegende of the Southern Alps in northern Italy.

The achieved results show that the whole sampled sequence is reversely magnetized, and magnetostratigraphic dating was therefore impossible. The Permian pole positions are in good agreement with the pole compilation of Krš (1978).

THE GEOLOGY OF THE THÜRINGER FOREST

The Th.F. represents a "Mittelgebirge", a low mountain range, with the highest peaks at about 900 m, situated in the southwest of the G.D.R.

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Geologically, it represents a Saxonian uplifted area with a width of about 15–20 km and a length of 70 km.

The strike is NW–SE. The southwestern and northeastern flanks are built by tectonically well defined edgefaults (flexures with upward and downward thrusting as well as overthrusting). On the southeastern side, the border of the Thüringer Schiefergebirge is mapped by the distribution of the "Rotliegende" and the Th.F. ends in the northwest because of the northward dipping below younger sediments.

In the Upper Carboniferous in the Thüringer area, basins and swells were formed, with the eroded sediments of the swell zones filling the basins. This morphological picture encompasses the terrain of the Th.F. and now forms the basis for its present tectonic structure. The Eisenacher basin can be seen as the relic of the former Saar–Selke basin. Further to the southeast the Ruhlaer Sattel is a relic of the Mitteldeutsche Kristallinschwelle, followed by the largest basin, the Oberhöfer basin, which is again a relic of the Saar–Selke basin (Beyschlag, 1895).

The series of the Rotliegende, which reaches a thickness of about 2000 m, consists of fluviatile and limnic sediments interbedded with volcanic rocks.

Geotectonically, the sediments are interpreted as molasses and volcanites as consequences of the Variscan orogenic cycle. The main mass of the rocks is light to dark red in colour. This red colour is generally interpreted as the result of high temperatures during the change from wet to dry periods starting in the area of erosion and continuing during transportation (Katzung, 1968).

SAMPLING

The sampling was carried out by drilling cores in the field and taking orientations with a magnetic or sun compass. Each site represents at least six cores and there are between three and eight sites in each locality. The lengths of the cores are around 7 cm in volcanites and up to 20 cm in sediments, in order to get unweathered samples.

ROCK MAGNETIC EXPERIMENTS

Because of the large variety of rocks, a great number of pilot specimens were selected to test the stability of the natural remanent magnetization (NRM), to find out the carrier of the remanence and whether the remanence is depositional (DRM) or chemical (CRM).

AC and thermal cleaning experiments as well as measurements of saturation were carried out and Is–T curves were plotted. The pilot specimens were partly measured in Potsdam and partly in the laboratory Gams of the Mining University Leoben. All these experiments established hematite as the main carrier of the NRM (Fig. 1 a-f). In some porphyrites magnetite can be observed as well as hematite (Fig. 1a, d, e). In the Manebacher coal-bearing formation, magnetite and possibly iron sulphide as the carriers of mineralization were found (Fig. 1a, b, c; TW 35, 36).

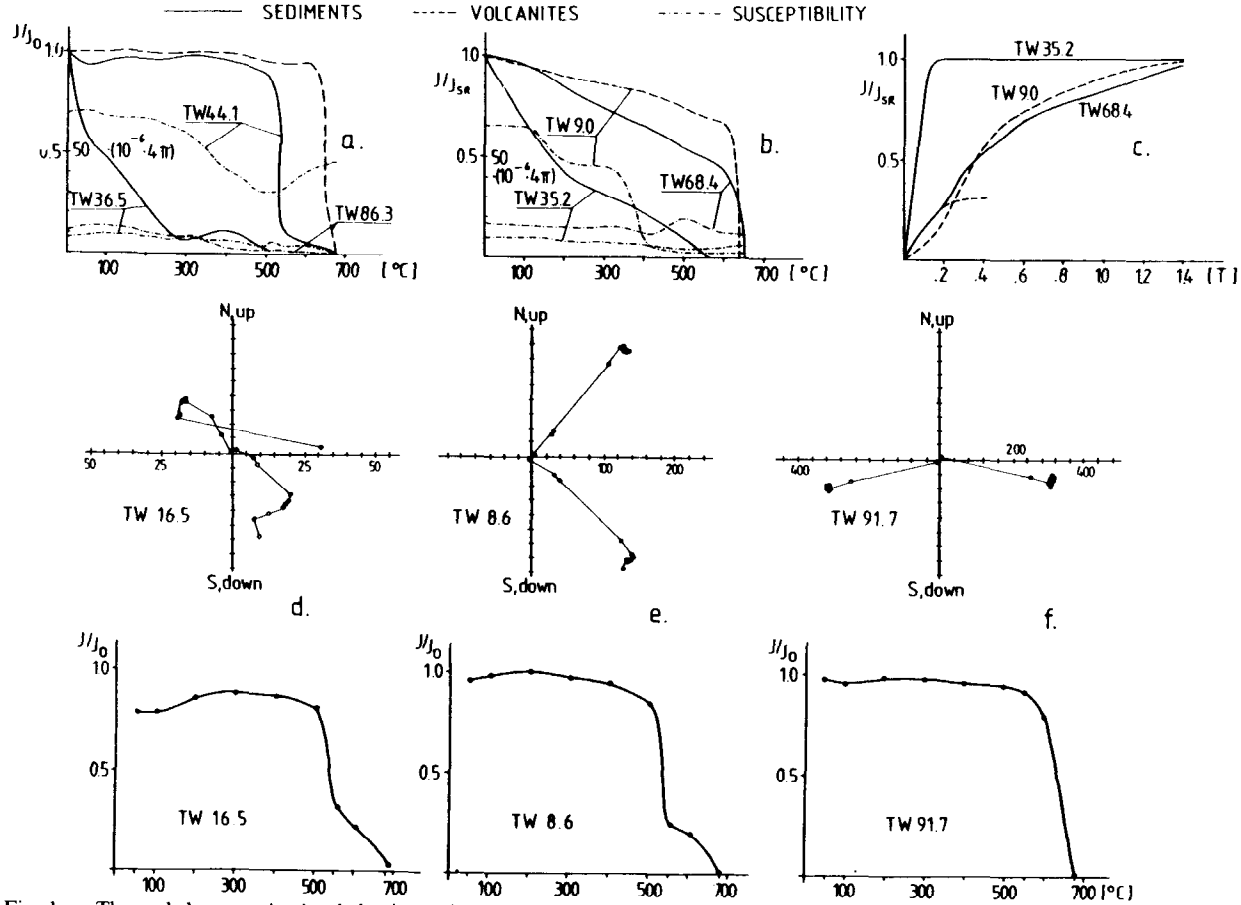


Fig. 1. a. Thermal demagnetization behaviour of sediment (TW 36.5, TW 44.1) and volcanite pilot specimens (TW 86.3). b. *Is*–*T*-curves of sediment and volcanite pilot specimens. c. Saturation aquisition curves. The dashed curve in TW 68.4 shows the effect of magnetite. d. Two-component mineralization (Fe_2O_3 and Fe_3O_4) with a strong effect of VRM in initial stage. e. Two-component mineralization (Fe_2O_3 and Fe_3O_4) where both show the same remanence direction. f. Single-component mineralization behaviour due to hematite.

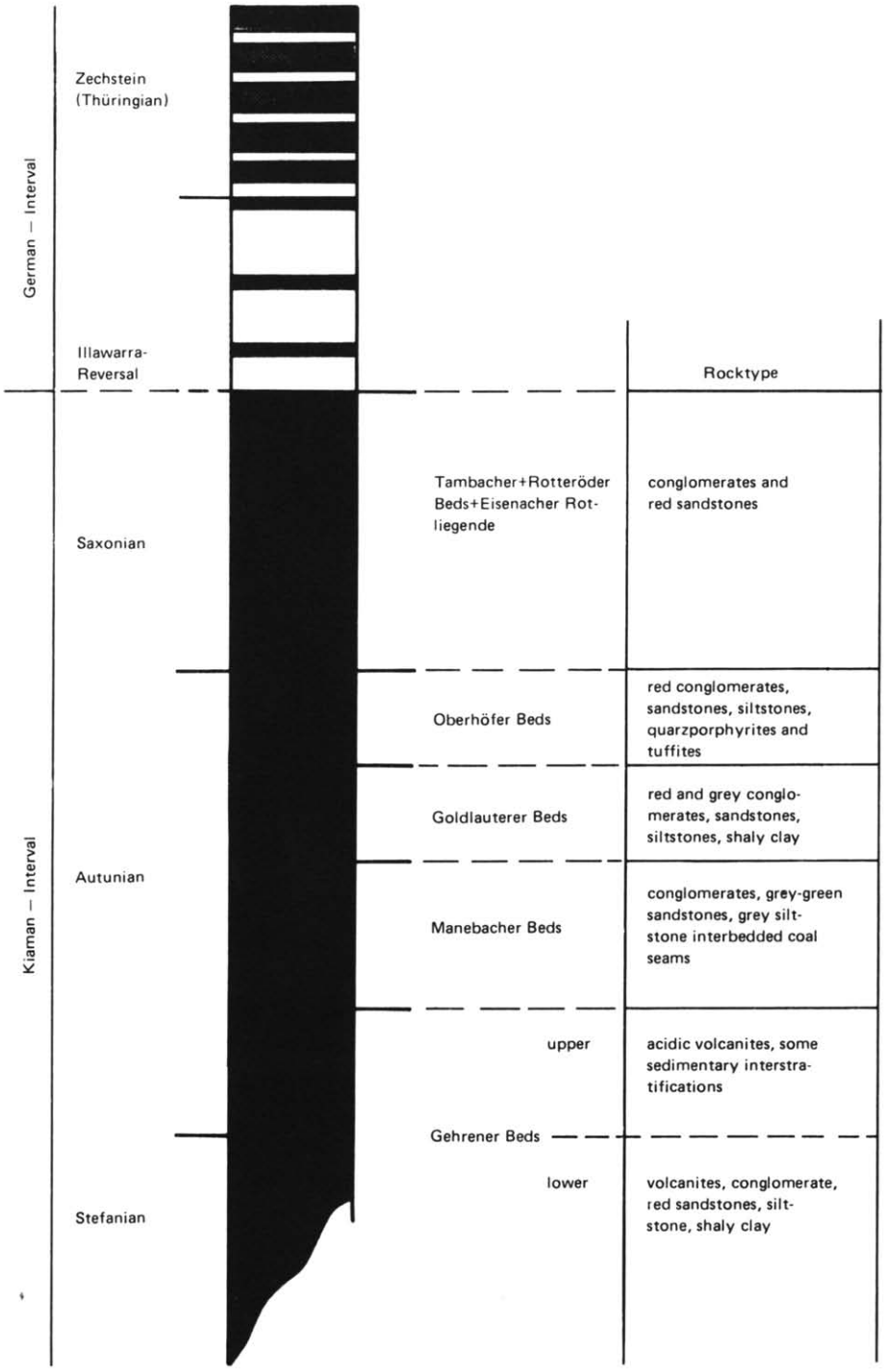


Fig. 2. Magnetostratigraphic time scale after Dachroth (1976) in comparison with the sampled sequences and rock types.

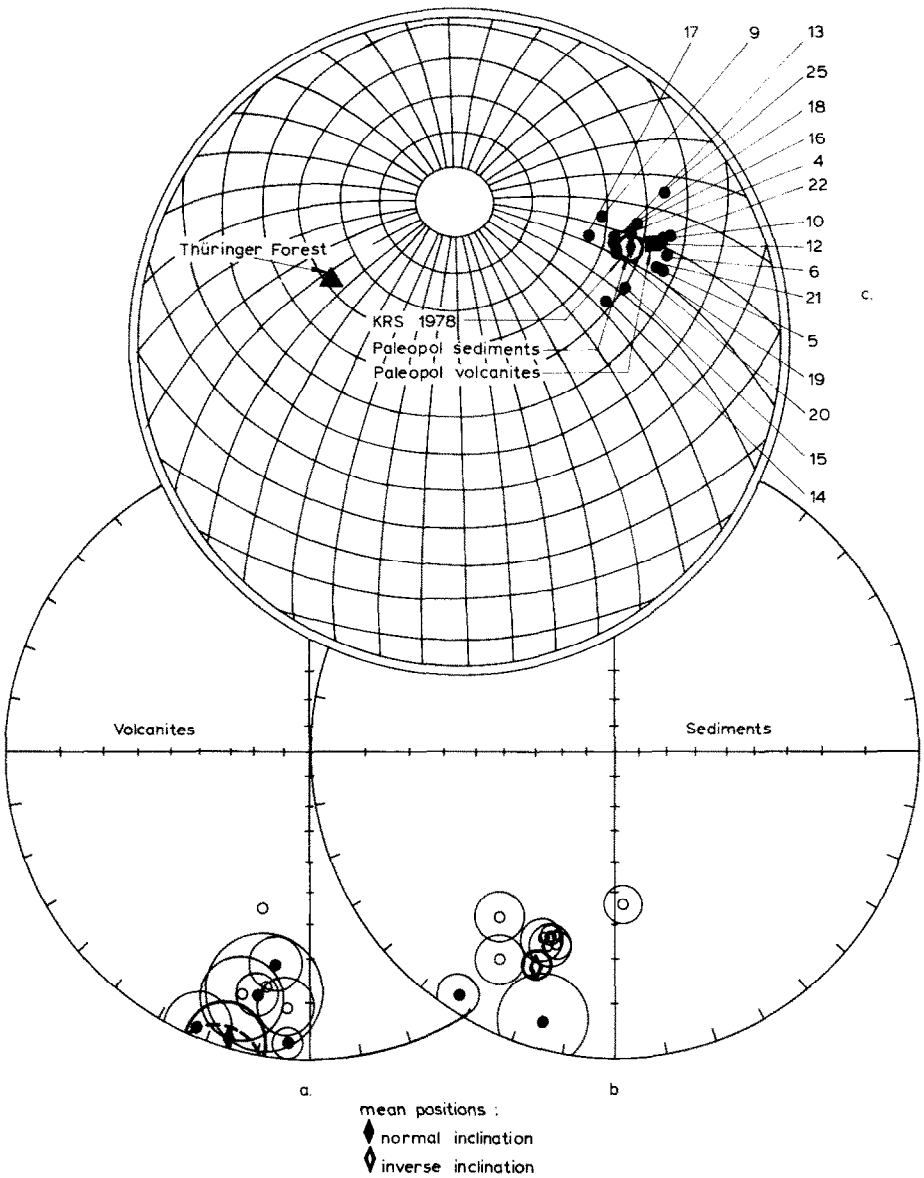


Fig. 3. a. Locality mean directions of the volcanites. b. Locality mean directions of the sediments. c. Calculated paleopole positions for the locality means and the overall pole position for volcanites and sediments.

The cleaning experiments proved the complete elimination of the weak secondary components of magnetization above 400°C. The main group of samples was cleaned at 400°C, the porphyrites which sometimes showed magnetite besides hematite were also cleaned at 550°C. In 20% of these samples, the result was remarkably better than after 400°C. The cleaned magnetization directions (ChRM) were treated after Fisher (1953).

TABLE I

Table of paleomagnetic results and poleposition of the Thüringer Forest

Site No.	Locality	Rock type	Stratigr. level	Paleo-decl.(°)	Paleo-incl.(°)
Tw 1– 3	Felsenkeller Ilmenau	"Sturmheide-porphyry"	up. Gehrener beds	232.1	−5.2
5– 8	150 m east of TW 1	up. porphyrite	up. Gehrener beds	233.1	27.6
9– 12	Hammergrund Ilmenau	"Höllkopf-melaphyre"	up. Gehrener beds	184.9	−7.2
13– 17	Str. Ilmenau-Stützerbach	mica-porphyrite	low. Gehrener beds	195.1	−10.2
18– 21	Kickelhahn	"Kickel-hahntuff"	up. Gehrener beds	190.4	−13.4
24– 25	Forestry Stützerbach	mica-porphyrite	low. Gehrener beds	198.2	6.0
26– 30	Station Stützerbach	quartzporphyry	low. Gehrener beds	212.8	39.6
31– 33	Str. Manebach-Stützerbach	low. porphyrite	low. Gehrener beds	151.5	−78.0
34– 36	Southern Manebach	sediments	Manebacher beds	not measurable	
37– 44	Höllkopf, N-flank	upper mudstones	up. Gehrener beds	186.1	−40.4
45– 52	Pochwerksgrd. Goldlauter	sandstone	low. Goldlauterer beds	195.7	−0.3
53– 57	Eisenach	red sediments	Eisenacher Rotleigendes	282.6	49.6
59– 60	Wartburg	"Wartburg-conglomerate"	Eisenacher Rotliegendes	192.3	41.6
63– 70	Tabarz	gr. a. viol sediments	Oberhöfer beds	184.8	−40.3
71– 74	Gasberg near Rotterode	red sediments	Rotteröder beds	207.1	7.6
75– 83	Nesselberg Tambach	red sediments	Oberhöfer beds	195.9	−20.1
84– 88	Tambach–Dietharz	young. quartz-porphyry	Oberhöfer beds	196.2	−32.6

Pole position		Polarity	α_{95}	k	Pole
lat.(°)	long.(°)		(°)		position nr.
−25.07	310.36	reversed	26.1	1.8	1
−9.89	319.12	reversed	46.3	4.8	2
−42.75	4.19	reversed	5.4	289.3	3
−42.72	350.21	reversed	9.5	64.9	4
−45.24	356.01	reversed	13.3	48.4	5
−34.04	348.68	reversed	91.0	1.5	6
−11.25	340.16	reversed	15.0	26.6	7
−68.42	160.56	reversed	24.3	26.5	8
−61.96	358.85	reversed	45.9	2.4	9
−37.73	350.86	reversed	7.9	49.0	10
30.80	292.63	reversed?	35.8	7.5	11
−14.55	359.22	reversed	91.0	2.0	12
−62.04	1.31	reversed	7.3	57.4	14
−37.85	335.61	reversed	9.0	103.0	15
−47.61	347.30	reversed	4.0	162.5	16
−54.58	343.52	reversed	1.3	1000.0	17

TABLE I (continued)

Site No.	Locality	Rock type	Stratigr. level	Paleo-decl.(°)	Paleo-incl.(°)
89– 91	Tambach–Dietharz	old. quartz-porphyry	Oberhöfer beds	189.5	18.4
92– 97	Tambach–Dietharz	red sandstone	Tambacher beds	198.8	– 17.1
98–108	Street Georgenthal	red sediments	Tambacher beds	199.7	– 14.9
109–116	Georgenthal	mica-porphyrite	low. Gehrener beds	202.5	2.1
117–121	Regenberg Friedrichroda	young. por-phyry	Oberhöfer beds	193.4	10.3
122–126	Friedrichroda	red sediments	up. Goldlau-terer beds	176.1	– 18.2
127–131	"Gottlob" Friedrichroda	red sediments	up. Goldlau-terer beds	178.1	– 32.7
132–135	Valley of small Leina	young. por-phyry	Oberhöfer beds	183.8	3.2
136–142	Station Oberhof	red sediments	Oberhöfer beds	212.8	– 7.8

RESULTS AND CONCLUSIONS

As far as the problem of depositional or chemical remanence is concerned, we found in all localities an increasing scatter of the directions of magnetization with increase in grain size, which seems to be a strong argument for DRM. The hematite pigment, responsible for the red colour of the rock matrix, occurs in a superparamagnetic particle size ($< 1\ \mu$) and is not the carrier of the remanent magnetization. These facts require two generations of hematite, one detrital the other chemical.

The polarity of the ChRM is all reversed (Fig. 2, Table I) with shallow inclinations. There is very good agreement between the locality means of both volcanites and sediments. This agreement and the high stability found during the cleaning experiments prove the suitability of the material and that the characteristic remanence (ChRM) is primary.

The data shown in Fig. 3a, b, and Table I were used to calculate the mean values of the declination and inclination for the Th.F. as well as the coordinates of the

Pole position		Polarity	α_{95}	k	Pole
lat.(°)	long.(°)		(°)		position nr.
−29.25	000.03	reversed	7.1	297.7	18
−45.27	343.89	reversed	11.1	36.9	19
−43.88	343.26	reversed	4.1	120.2	20
−34.82	343.39	reversed	6.3	76.2	21
−32.86	354.92	reversed	4.1	333.9	22
−48.53	16.66	reversed	8.3	122.1	23
−57.09	14.06	reversed	46.0	3.7	24
−37.59	5.97	reversed	3.7	602.1	25
−35.73	329.12	reversed	5.0	143.6	26

paleopole of this area (Table II). Localities with $\alpha_{95} > 20°$ were not accepted for this calculation.

The calculated pole positions show a perfect fit with the Permian pole in the compilation by Krš (1978). The above-mentioned suspicion that the Illawarra Reversal should occur in the Tambacher, Rotteröder and/or the Eisenacher Rotliegende beds was not proved. The whole sampled sequence was reversed.

TABLE II

Table of final results *

	Lat.(°)	Long.(°)	Paleodecl.(°)	Paleoincl.(°)	k	α_{95}	P_{lat}	P_{long}
Sediments	50,70	10,90	199,3	−19,1	9,7	5,8	41,54	159,86
Volcanites	50,70	10,90	196,6	+0,5	9,7	7,1	37,11	169,87

* In this table P_{lat} and P_{long} are the coordinates of the paleopole.

This leads to the conclusion that even the highest sequences of the Rotliegende in the Th.F. are older than the Illawarra Reversal.

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